Table of Fourier Transform Pairs

Property	**Signal**	**FT in f**	**FT in ω**
Linearity	$ax_1(t) + bx_2(t)$	$aX_1(f) + bX_2(f)$	$aX_1(\omega) + bX_2(\omega)$
Time delay	$x(t - t_0)$	$X(f)e^{-j2\pi f t_0}$	$X(\omega)e^{-j\omega t_0}$
Frequency Translation	$x(t)e^{j2\pi f_0 t}$	$X(f - f_0)$	$X(\omega - \omega_0)$
Convolution	$x_1(t) * x_2(t)$	$X_1(f)X_2(f)$	$X_1(\omega)X_2(\omega)$
Multiplication	$x_1(t)\, x_2(t)$	$X_1(f) * X_2(f)$	$\frac{1}{2\pi} X_1(\omega) * X_2(\omega)$
Parseval's Theorem	$\int_{-\infty}^{\infty} \lvert x(t)\rvert^2 dt$	$\int_{-\infty}^{\infty} \lvert X(f)\rvert^2 df$	$\frac{1}{2\pi} \int_{-\infty}^{\infty} \lvert X(\omega)\rvert^2 d\omega$
Rectangle	$\Pi\left(\frac{t}{\tau}\right)$	$\tau\, \mathrm{sinc}(f\tau)$	$\tau\, \mathrm{sinc}\left(\frac{\omega\tau}{2\pi}\right)$
sinc()	$\mathrm{sinc}(Wt)$	$\frac{1}{W}\Pi\left(\frac{f}{W}\right)$	$\frac{1}{W}\Pi\left(\frac{\omega}{2\pi W}\right)$
$\mathrm{sinc}^2()$	$\mathrm{sinc}^2(Wt)$	$\frac{1}{W}\,\mathrm{tri}\left(\frac{f}{W}\right)$	$\frac{1}{W}\,\mathrm{tri}\left(\frac{\omega}{2\pi W}\right)$
Triangle	$\Lambda\left(\frac{t}{\tau}\right)$	$\tau\, \mathrm{sinc}^2(f\tau)$	$\tau\, \mathrm{sinc}^2\left(\frac{\omega\tau}{2\pi}\right)$
Exponential	$e^{-at}u(t),\ a > 0$	$\frac{1}{a+j2\pi f}$	$\frac{1}{a+j\omega}$
Impulse	$A\delta(t)$	A	A
Constant	A	$A\delta(f)$	$2\pi A\delta(\omega)$
Complex exponential	$e^{j2\pi f_0 t}$	$\delta(f - f_0)$	$2\pi\delta(\omega - \omega_0)$

Energy Signals

Signal	$x(t)$	**FT in f**	**FT in ω**
Impulse	$\delta(t)$	1	1
DC (constant)	1	$\delta(f)$	$2\pi\delta(\omega)$
Cosine	$\cos(2\pi f_0 t)$	$\frac{1}{2}\left[\delta(f - f_0) + \delta(f + f_0)\right]$	$\pi\left[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)\right]$
Sine	$\sin(2\pi f_0 t)$	$\frac{1}{2j}\left[\delta(f - f_0) - \delta(f + f_0)\right]$	$j\pi\left[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)\right]$
Complex exponential	$e^{j2\pi f_0 t}$	$\delta(f - f_0)$	$2\pi\delta(\omega - \omega_0)$
Unit step	$u(t)$	$\frac{1}{j2\pi f} + \frac{1}{2}\delta(f)$	$\frac{1}{j\omega} + \pi\delta(\omega)$
Signum	$\mathrm{sgn}(t)$	$\frac{1}{j\pi f}$	$\frac{2}{j\omega}$